

Fig. 3: Spectral profile of an IR emission from a typical target aircraft

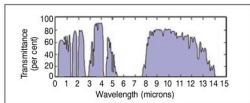


Fig. 4: Transmission characteristics of the atmosphere

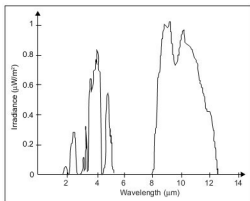


Fig. 5: IR spectrum as seen by the seeker head of an IR-guided missile

niques, the target shape can be used to find its most vulnerable part. This information can then be used to steer the weapon towards the target.

Non-IIR-guided missiles make use of IR emission corresponding to thermal signatures of the exhaust and the mainframe of the target aircraft to home on it. Emission in 3-5 and 8-12 micron bands is characteristic of electromagnetic emission from jet exhaust and mainframe of the aircraft. Fig. 3 shows the spectral profile of IR emission from different parts of a typical target aircraft. Spectral content of IR emission as received

by a seeker head is the superposition of spectral emission of the aircraft on the transmission characteristics of the atmosphere (Fig. 4).

Fig. 5 shows a typical IR-emission spectrum that would be seen by the non-imaging seeker head. This wavelength signature is judiciously used in guidance of air-to-air and surface-to-air IR-guided missiles.

Modern IR seekers also operate in the 8-12 micrometre wavelength range, which is absorbed least by the atmosphere. Such seekers are called two-colour systems. Two-colour seekers are harder to defeat with countermeasures such as flares and jammers.

IR-guided missiles developed in 1970s and 1980s used single-colour IR seekers employing 3-5 micron band. IR seekers used in these missiles were most effective in detecting IR radiation of shorter wavelengths such as 4.2 micron emission of carbon dioxide efflux of a jet engine. Seekers responding to 3-5 micron band were called single-colour seekers and missiles as single-colour missiles.

MAGIC series air-to-air missiles from France and R-73 air-to-air missiles from Russia are some examples. State-of-the-art IR-guided missiles use seekers that respond to both 3-5 and 8-12 micron bands to offer improved false alarm rejection and immunity to deception by flares. These seekers are called two-colour seekers and missiles, two-colour missiles. Python from Israel and RVVAE from Russia are examples of missiles using two-colour seeker heads.

Also, both surface-to-air and air-to-air IR-guided missiles receive target's IR signatures in the presence of background radiation from the sky and also IR signatures of flares, if any, deployed by the target aircraft platform. The seeker head should be able to discriminate between IR signatures

of the background and flares from those of the target.

Advantages and limitations

IR-guided missiles and radar-guided missiles are deployed in similar roles, predominantly as surface-to-air and air-to-air guided weapons. It would therefore be nothing but logical to compare the two types of guided weapons.

IR-guided missiles offer several advantages over radar-guided missiles. One, IR-guided missiles are far more immune to electronic countermeasures than their radar counterparts.

Two, IR-guided missiles offer greater safety of pilots of the aircraft carrying these weapons. Due to the inherent fire-and-forget capabilities of IR-guided missiles, after releasing the missile, the pilot can leave the area while the missile guides itself to the target.

Three, IR-guided missiles are almost impossible to detect during launch preparation. On the other hand, radar-guided missiles employ either beam rider or semi-active or active guidance. In all forms, it employs a radar, which can be detected through radar emission during launch preparation or flight.

Four, IR-guided missiles are manoeuvring missiles and are particularly suitable for close engagement.

Five, IR sensors perform well during day and night conditions.

Six, IR sensors employed by heat-seeking missiles cost less per unit.

While radar-guided missiles are all-weather weapons and have relatively much longer operational ranges, heat-seeking missiles perform better in close-in ranges. Also, heat-seeking missiles are vulnerable to the use of passive and active IR countermeasures. Use of chaff, decoys and flares to deceive heat-seeking missiles is a common occurrence. Heat-seeking missiles employing IIR seekers though are far less vulnerable to these countermeasures.

In the early days of development of heat-seeking missiles, these missiles could only lock on to intense sources of heat such as jet exhaust pipes, and therefore could be fired only from be-